Zeeshan Manzoor Bhat

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RESEARCH INTERESTS

Large-scale experimental testing; Nonlinear modelling of structures; Seismic hazard, risk and vulnerability assessment; Performance-based earthquake engineering; Resilience-based seismic design.

EDUCATION

Ph.D. in Structural and Earthquake Engineering *July 2018 – March 2025*
Indian Institute of Technology Roorkee, India
Thesis: *Seismic Performance of Masonry Infilled RC Frame Buildings Designed for Modern Codes*
Advisors: Prof. Yogendra Singh & Prof. Pankaj Agarwal

M.Tech. in Structural Engineering *July 2016 – July 2018*
National Institute of Technology Srinagar, India
Thesis: *An Experimental Investigation into the Behavior of Steel–Timber Composite Beams*
Advisor: Prof. Javed Ahmad Bhat

B.Tech. in Civil Engineering *July 2012 – June 2016*
National Institute of Technology Srinagar, India

RESEARCH EXPERIENCE

CITRA Postdoctoral Fellow *August 2026 – Ongoing*
Queen's University, Kingston, Canada
Project: *Next-Generation Climate-Resilient and Energy-Efficient Housing*

Research Associate *June 2025 – June 2026*
Indian Institute of Technology Roorkee, India
Project: *Enhancing Sustainability of RC Frame Buildings through Seismic Safety and Design Life Elongation Using GFRP Reinforcement*

FUNDED PROJECTS

Next-Generation Climate-Resilient and Energy-Efficient Housing
Queen's University, Kingston, Canada
Funding: $140,000 through Canada Impact + Research Training Awards
Supervisor: Prof. Amir Fam

PUBLICATIONS

Peer-Reviewed Journal Articles

[1] **Bhat, Z. M.**, & Singh, Y. (2026). Out-of-plane seismic fragility analysis of masonry infills in low- and mid-rise RC frame archetype buildings accounting for in-plane damage. ***Journal of Building Engineering.*** Link

[2] **Bhat, Z. M.**, & Singh, Y. (2025). Seismic risk assessment of fly ash brick and AAC block masonry infilled RC frame buildings designed for modern codes. ***Journal of Earthquake Engineering.*** Link

[3] **Bhat, Z. M.**, & Singh, Y. (2025). Out-of-plane behavior of masonry infills of different types and slenderness ratios under reversed cyclic loading. ***Journal of Structural Engineering.*** Link

[4] **Bhat, Z. M.**, Singh, Y., & Agarwal, P. (2024). Cyclic testing and diagonal strut modelling of different types of masonry infills in reinforced concrete frames designed for modern codes. ***Engineering Structures.*** Link

[5] **Bhat, Z. M.**, Singh, Y., & Agarwal, P. (2023). Characterization of mechanical behavior of different types of masonry with full-field strain using digital image correlation. ***Construction and Building Materials.*** Link

[6] Waseem, S. A., **Bhat, Z. M.**, & Bhat, J. A. (2022). Experimental investigation into the behavior of steel–timber composite beams. ***Practice Periodical on Structural Design and Construction.*** Link

Under Review

[7] **Bhat, Z. M.**, & Singh, Y. Full-scale cyclic testing and DIC-based damage assessment of masonry infilled RC frames with openings. ***ACI Structural Journal.***

INTERNATIONAL CONFERENCES

[1] **Bhat, Z. M.**, & Singh, Y. (2025). Reversed out-of-plane cyclic testing of masonry infills using double-leaf airbags. ***COMPDYN 2025: 10th International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering***, Rhodes Island, Greece, 15–18 June 2025. Link

[2] **Bhat, Z. M.**, & Singh, Y. (2025). Cyclic in-plane behavior of masonry infilled RC frames with detailed interpretation of damage using digital image correlation. ***15th Canadian Masonry Symposium***, Ottawa, Canada, 2–5 June 2025. Link

TECHNICAL REPORTS

[1] **Bhat, Z. M.**, Singh, Y., & Agarwal, P. (2024). *Seismic Safety Measures for RC Frame Buildings with Different Types of Infill Panels.* NBCC (India) Limited, India.

[2] Team Member. (2022). *Compendium of Traditional Earthquake-Resilient Construction Practices.* NDMA, Government of India. Link

OPEN DATASETS

[1] **Bhat, Z. M.**, & Singh, Y. (2024). Fly ash brick and AAC block masonry infilled RC frame buildings: OpenSees models and responses. ***DesignSafe-CI.*** Link

AWARDS AND ACHIEVEMENTS

- Canada Impact + Research Training Award – Postdoctoral, **Natural Sciences and Engineering Research Council of Canada (NSERC)**, 2026
- Research Fellowship, **Sapienza University of Rome, Italy**, 2025
- MHRD Fellowship for Doctoral Studies, **Ministry of Human Resource Development, Government of India**
- MHRD Fellowship for M.Tech Studies, **Ministry of Human Resource Development, Government of India**
- Ranked among the top 2% in the **Graduate Aptitude Test in Engineering (GATE)** 2016, among 119,873 candidates

TECHNICAL SKILLS

- **Programming & Tools:** Python, MATLAB, TCL, LaTeX
- **Software:** OpenSees, STKO, ETABS, SAP2000, OpenQuake
- **Experimental Techniques:** Digital Image Correlation (DIC), Laboratory Instrumentation and Data Acquisition, Servo-hydraulic Actuator Control, Load Cell Calibration

INTERNATIONAL WORKSHOPS

6th International Course on Seismic Analysis of Structures — *19–22 July 2021*
University of Palermo, Italy

Nonlinear Modelling of RC Structures Using OpenSees — *22–26 June 2020*
Università degli Studi di Napoli Federico II, Italy

TEACHING ASSISTANTSHIPS

Advanced Earthquake Resistant Design of Structures (EQN-512) — *IIT Roorkee*
Instructor: Prof. Yogendra Singh

Earthquake Resistant Design of Structures (EQN-563) — *IIT Roorkee*
Instructor: Prof. Yogendra Singh

REFERENCES

Prof. Yogendra Singh
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